

Document on observability

Observability Metrics:

- Observability metrics are numerical data that help monitor the performance and health of applications, systems, and infrastructure. These metrics provide insights into how a system behaves in real time and help engineers detect problems, troubleshoot issues, and improve system reliability.
- Observability mainly uses three types of telemetry data: **metrics, logs, and traces**, where metrics represent measurable values such as CPU usage, request count, response time, and error rates.

Three Pillars of Observability

1. Metrics

Metrics are numerical measurements that represent the performance and health of a system over time. They are collected at regular intervals and help monitor system behavior through graphs and dashboards.

Examples:

- CPU usage
- Memory utilization
- Request rate
- Error rate
- Response time

Metrics are useful for **monitoring trends, detecting anomalies, and triggering alerts** when system performance crosses predefined thresholds.

2. Logs

Logs are detailed records of events that occur within an application or system. They capture information about what happened, when it happened, and sometimes why it happened.

Examples:

- Error messages
- System warnings
- Application events
- Debug information

Logs are mainly used for **troubleshooting and debugging issues** because they provide detailed context about system events.

3. Traces

Traces track the complete journey of a request as it moves through different services in a distributed system. Each step of the request is recorded, helping engineers understand how different components interact.

Examples:

- API request flow through microservices
- Database query execution path
- Service-to-service communication

Traces help identify **performance bottlenecks, latency issues, and dependencies between services**.

Observability metrics are commonly categorized into three types:

- **Application metrics** – measure application performance such as request latency, response time, and error rate.
- **Infrastructure metrics** – monitor system resources like CPU usage, memory consumption, disk I/O, and network traffic.
- **Business metrics** – track business-related activities such as number of users, transactions, or revenue generated.

What is observability?

The ability to understand what is happening inside a system by looking at its outputs.

Why observability?

It helps to find the root cause fast and improves reliability.

Golden Signals:

In Site Reliability Engineering (SRE), thousands of metrics can exist, but focusing on the most important ones helps simplify monitoring. The **Golden Signals** are four key metrics used to measure system reliability and performance.

1. Latency

Latency measures the **time taken for a system to respond to a request**. High latency indicates slow system performance and poor user experience.

2. Traffic

Traffic represents the **amount of demand on a system**, usually measured as requests per second or number of transactions.

3. Errors

Errors measure the **rate of failed requests or system failures**, such as HTTP errors or application exceptions.

4. Saturation

Saturation indicates **how close a system is to reaching its resource limits**, such as high CPU usage, memory consumption, or queue length.